NATALY R. PANCZYK

npanczyk@umich.edu | 847-767-2034

EDUCATION

University of Michigan — **In-Progress**
PhD in Nuclear Engineering and Radiological Sciences
Research Focus: Validation and Interpretation Methods for Safety-Critical AI
GPA: 4.0/4.0

University of Illinois at Urbana-Champaign — **May 2024**
Bachelor of Science in Nuclear, Plasma, and Radiological Engineering (NPRE)
Concentration: Power, Safety, and the Environment
GPA: 3.94/4.0 Minor: Political Science

EXPERIENCE

Guest Researcher – *Los Alamos National Laboratory* — **June 2025 – April 2026**

- Developed an automatic neutron resonance detection framework using a fully convolutional network
- Applied deep learning methods to advance nuclear data analysis capabilities

TA NPRE 455 *(Neutron Diffusion and Transport) – University of Illinois* — **Jan. 2024 – May 2024**

- Hosted office hours; graded problem sets and computer projects

Intern – *MPR Associates* — **May 2023 – July 2023**

- Analyzed a reactivity insertion accident for USNC Micro Modular Reactor using RELAP5-3D
- Developed Python-based, maintenance dose tool for a private fusion system

Research Assistant – *Multiphase Thermo-fluid Dynamics Laboratory – UIUC* — **Jan. 2023 – May 2024**

- Analyzed post-CHF conditions for downward flow boiling in a vertical square channel

Research Assistant – *Advanced Reactors and Fuel Cycles – University of Illinois* — **Sept. 2020 – Dec. 2022**

- Began TRISO fuel analysis with MOOSE for the UIUC microreactor project
- Designed techno-economic analyses for clean hydrogen systems

Delegate – *Nuclear Engineering Student Delegation (NESD)* — **Sept. 2022, Sept. 2023**

- Presented policy proposals to various organizations (DOE, NEI, NRC, Thirdway, Clearpath, etc.) and Congressional representatives in D.C.

Intern – *Pacific Northwest National Laboratory* — **June 2022 – August 2022**

- Automated nuclear waste management modeling quality assurance and data visualization techniques

Intern – *Idaho National Laboratory* — **May 2021 – August 2021**

- Conducted a technoeconomic analysis on the feasibility of implementing microreactors into a clean steel manufacturing scheme using Temoa, an opensource energy system optimization model

AWARDS

National Science Foundation Graduate Research Fellowship — **April 2024**
University Nuclear Leadership Program Graduate Fellowship *(Declined)* — **April 2024**
NPRE Outstanding Academic Achievement Award to a Graduating Senior — **March 2024**
Kirkwood Women in Engineering Scholarship — **September 2023**
Roy G. Post Waste Management Scholarship — **December 2022**
Nuclear Energy Undergraduate Program (NEUP) Scholarship — **April 2022, April 2023**

SKILLS

Python (PyTorch, TensorFlow, SciPy), Git, LATEX, Working fluency in Spanish